Coypu adaptive management

Adaptive management

Decision theoretic methods can be used to make control decisions not only in the face of uncertainty, but also to reduce uncertainty over time. Adaptive management, or "learning by doing", involves making decisions to achieve a management objective while simultaneously learning about system dynamics to improve future management success (Holling 1978; Walters 1986). Adaptive management approaches differ based on whether they learn only from the past (passive adaptive management) or anticipate future learning (active adaptive management) (Chadès et al. 2017).

Here we apply adaptive management strategies to inform control decisions for a single coypu population and to learn about and reduce uncertainty in the per-capita growth rate. We will use the following discrete-time population model, where r is the intrinsic rate of growth; K is the carrying capacity; $q(u)$ is the harvest rate as a function of action u; R is the per-capita growth rate; m describes whether the per-capita growth rate is linear, concave, or convex; and ϵ is log-normally distributed noise:

$$N_{t+1} = N_t(1 + R(N_t)) - q(u) \times N + \epsilon$$

$$R(N_t) = r \times (1 - (N_t/K)^m)$$

If $m > 1$, most density—dependent change occurs at high population levels (close to the carrying capacity), which is theoretically common for species with life history strategies typical of large mammals. If $m < 1$, most density-dependent change occurs at low population levels, which is theoretically common for species with life history strategies typical of insects and some fishes (Fowler 1981).

We will consider a model set with five possible values of m. This model bounds, but does not contain, the true model we will use to simulate dynamics in the passive adaptive management approach (Figure 1A). These different values of m result in different optimal policies when system dynamics are known perfectly (Figure 1B).

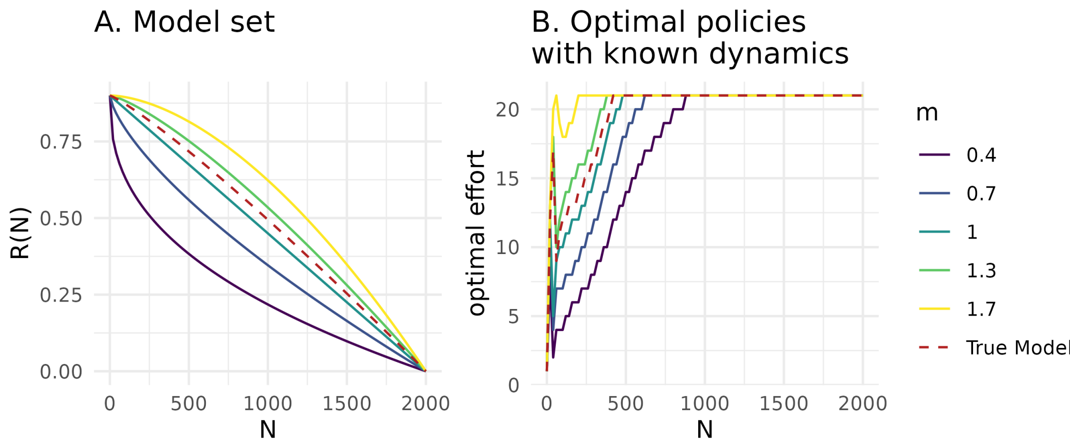

Figure 1: *A*. Model set representing structural uncertainty in the per-capita growth rate. *B*. Optimal state-dependent removal actions assuming perfectly known population dynamics, solved with stochastic dynamic programming. In both panels, colors correpond to different values of m in the model set, and the red dashed line corresponds to the true model dynamics used for simulation in the passive adaptive management approach.

Value of Information

For a decision maker, uncertainty may not be important to resolve; learning may not be inherently valuable if it does not result in improved management outcomes. A Value of Information (VoI) analysis *a priori* quantifies the importance of reducing uncertainty, or the expected improvement on management objectives if uncertainty were resolved (Raiffa and Schlaifer 2000). If the VOI is high, an adaptive management approach is typically justified.

The expected value of perfect information (EVPI) is measured as the difference between 1) PI, or the expected value once uncertainty has been resolved because the optimal action is chosen after knowing which model best describes the system, and 2) NL, or the expected value in the face of uncertainty because the optimal action is chosen with no learning about the system (Runge, Converse, and Lyons 2011). The difference between these terms, EVPI $= PI - NL$, is the expected improvement in management performance due to acquiring perfect information.

To calculate EVPI for a sequential decision problem, we first use SDP to calculate the optimal state-dependent policy with assumed system dynamics, $\pi^*(s)_k$, for each model k in the model set K, given each transition matrix, P_k. We also calculate the bet-hedging strategy, or the optimal policy given uncertain dynamics, $\pi^*(s)_{NL}$. This bet-hedging strategy is calculated with the transition matrix P_{NL}, representing the average of the transitions of each model, P_k, weighted by each model belief, b: $P_{NL}(s_{t+1}|s_t, a_t) =$

$\sum_{k \in K} b(k) P_k(s_{t+1}|s_t, a_t)$.

Using the policies calculated with known dynamics, $\pi^*(s)_k$, and unknown dynamics, $\pi^*(s)_{NL}$, we find the EVPI by first simulating i replicated population trajectories assuming each model k, applying both policies with known and unknown dynamics, and calculating the value at time t over time horizon T. The value associated with applying the policy with known dynamics is $V_{k,t,i}(s_t, a_t|a_t = \pi^*(s)_k)$, and the value associated with applying unknown dynamics is $V_{k,t,i}(s_t, a_t|a_t = \pi^*(s)_{NL})$. The EVPI is therefore $E_{s,k,t,i}[V(s_t, a_t|a_t = \pi^*(s)_k)] - E_{s,k,t,i}[V(s_t, a_t|a_t = \pi^*(s)_{NL})]$.

In the context of coypu regulation, the EVPI represents the expected improvement on the management objective if uncertainty about the per-capita growth rate was reduced. We calculate the EVPI for a model set, K, where $m \in \{0.4, 0.7, 1, 1.3, 1.7\}$ for $I = 200$ iterations over a $T = 40$ time horizon. We assume equal belief in each model. The expected value once uncertainty has been resolved, PI, is -1730.4, the expected value with no learning, NL, is -1787.8, and the EVPI is $PI - NL = 57.4$. It is therefore valuable to learn about the population dynamics and per-capita growth rate, as this learning is expected to improve performance on the management objective.

Passive adaptive management

Passive adaptive management involves learning from the outcomes of previous management actions to update knowledge about system dynamics and improve management performance over time. Decisions at each time step are optimized assuming that current knowledge of the system will not change into the future. Learning therefore occurs by monitoring the effect of an implemented action, rather than during the optimization procedure (Chadès et al. 2017).

Solving a passive adaptive management problem includes an implementation step and an updating step at every time point t, based on the current belief in model k, $b_t(k)$. The MDP must be solved at every time step because the transition matrices change as the belief changes. During the implementation step, the optimal action at time t is selected using a weighted averaging approach, where the transition probabilities are averaged across all models with the weights given by the current belief, $b_t(k)$ (Williams, Eaton, and Breininger 2011). This can be solved using stochastic dynamic programming:

$$V^*(s_t|b_t) = \max_{a \in A} \left\{ \sum_{k \in K} b_t(k) r(s_t, a_t) + \gamma \sum_{s_{t+1}} \left\{ \sum_{k \in K} b_t(k) P_k(s_{t+1}|s_t, a_t) \times V^*(s_{t+1}|b_t) \right\} \right\}$$

During the updating step, the resulting state, s_{t+1}, after taking action a_t is observed, and belief, b_{t+1}, is updated according to Bayes' rule:

$$b_{t+1}(k|s_t, a_t, s_{t+1}, b_t) = \frac{P_k(s_{t+1}|s_t, a_t)b_t(k)}{\sum_{k\in K} P_k(s_{t+1}|s_t, a_t)b_t(k)}$$

For coypu control decisions, we solve the passive adaptive management problem by updating the belief about the coypu population dynamics to improve management performance over time. We begin with an equal belief, $b_{t=1}(k) = 1/5$, across the five models in the model set (Figure 1A). We iterate between 1) calculating the optimal action given the current belief state, 2) implementing the optimal action and simulating the next state, s_{t+1}, given the true population dynamics, and 3) updating the belief state based observing the next state given action a_t. Here we assume the state is perfectly observed. We simulate the application of passive adaptive management over 30 time steps. The belief in each model changes over time (Figure 2A), resulting in the highest belief for the two models closest to the true population dynamics $m = 1$ and $m = 1.3$ (Figure 1A). Through updating the belief state over time and implementing the optimal action given the current belief state, the management performance improves over time (Figure 2B).

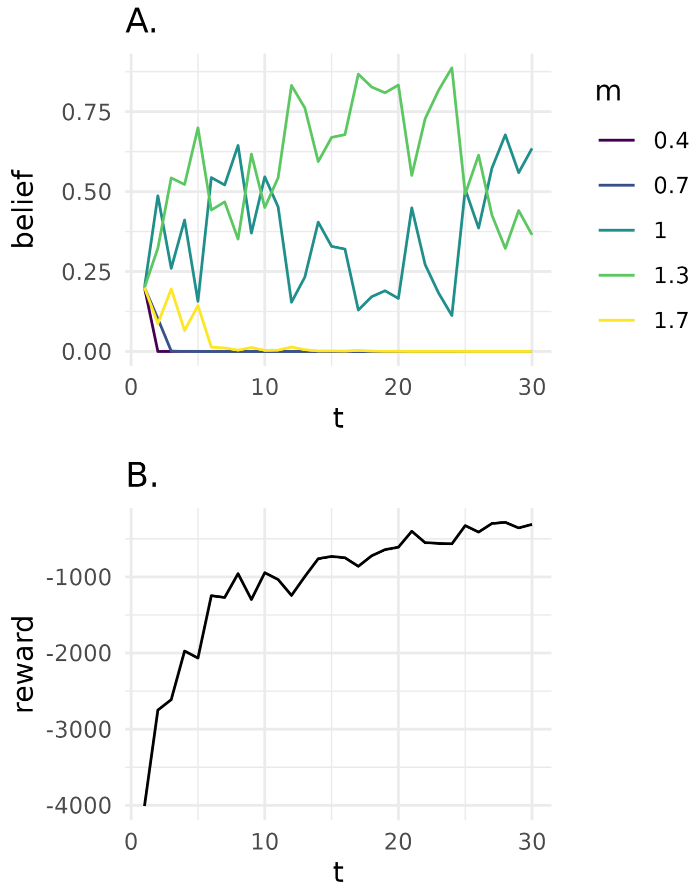

Figure 2: *A*. Belief state for each model k at time t. Colors correspond to different values of m, representing structural uncertainty in the per-capita growth rate. *B*. Reward, $r(s_t, a_t)$, at time t in passive adaptive management problem.

Active adaptive management

In contrast to passive adaptive management, active adaptive management does not rely only on past experience and instead explicitly accounts for future learning opportunities to maximize the chance of achieving the objective. Instead of an optimization at every time step as in passive adaptive management, the optimization occurs once and requires that the trajectory of belief b_t to b_{t+1} and state transitions be calculated for all time steps (Chadès et al. 2017). The optimal value function V^* that characterizes the performance of a policy is a function of both the state of the system s_t and belief over the models b_t:

$$V^*(s_t, b_t) = \max_{a \in A} \left\{ \sum_{k \in K} b_t(k) r(s_t, a_t) + \gamma \sum_{s_{t+1}} \left\{ \sum_{k \in K} b_t(k) P_k(s_{t+1} | s_t, a_t) \times V^*(s_{t+1}, b_t) \right\} \right\}$$

Active adaptive management with model uncertainty can be solved with many methods, including those developed for dealing with partial observability (Chadès et al. 2017; Williams, Eaton, and Breininger 2011). Here, a POMDP is characterized by a tuple $<X, A, O, P, Z, r, \gamma>$, where $X = S \times K$ represents the factored state space with S denoting the possible conditions of the system and K denoting the model set. The unknown model is an unobserved state variable that must be inferred through observation. We can solve the POMDP with the Successive Approximations of the Reachable Space under Optimal Policies (SARSOP) algorithm (Kurniawati et al. 2008). With this algorithm, the value function V^* associated with the optimal policy π^* can be approximated by a convex, piecewise-linear function $V(b) = \max_{\alpha \in \Gamma}(\alpha \cdot b)$, where Γ is a finite set of α-vectors and b is the discrete vector representation of a belief (Kurniawati et al. 2008).

For coypu control decisions, we solve the active adaptive management problem by augmenting the state space with the belief in two models where $m \in \{0.7, 1.3\}$. We choose a smaller model set than in the passive adaptive management problem due to the computational cost associated with exploring the belief and state spaces over the entire time period (i.e., "curse of dimensionality"). We assume a uniform belief across models and assume perfect observation of the state (i.e., $o_t = s_t$).

We also use the coypu example to demonstrate the "dual control problem" (Walters and Hilborn 1978). A management action altering a poorly understood system can produce two types of benefits: 1) short-term payoffs and 2) learning about the system so as to improve payoffs in the long-term. There is a trade-off between these two benefits, as large disturbances allow for rapid learning yet may incur high costs; the challenge is to optimally balance these benefits. Here we explore the dual control problem by selecting two different discount factors, $\gamma \in \{0.95, 0.75\}$. The higher of which represents a decision maker who values long-term rewards more highly, and the lower of which represents a decision maker who values short-term rewards more highly.

The active adaptive management solution describes the optimal action (a_t, removal effort), given the state (s_t, coypu abundance), initial belief ($b_{t=1}$, uniform across models), and discount factor (Figure 3). The solution differs depending on how future rewards are discounted in the optimization. At low coypu abundance, the optimal removal effort increases as the discount factor increases. With a higher discount factor, future system states are valued more highly; even though a higher removal effort is costly today, this higher removal effort will accelerate learning to yield higher rewards in the long term (Figure 3A). In contrast, with a lower discount factor, active learning becomes less useful, as the price paid for not following the passive-adaptive (optimal right now) solution exceeds the information gained from learning (Figure 3B).

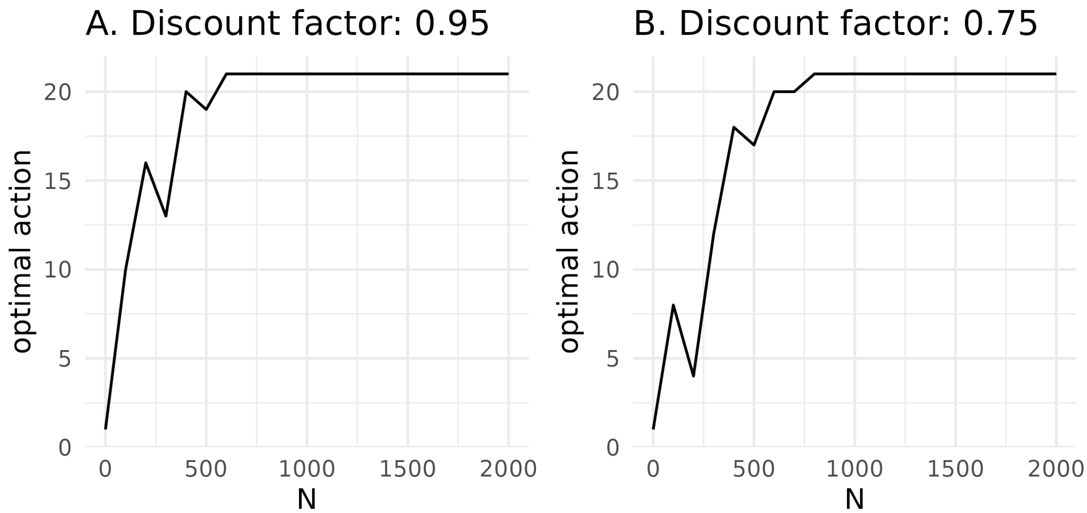

Figure 3: *A*. State-dependent active adaptive management solution, given an initial uniform belief across models and a higher discount factor representing a decision maker who values long-term payoffs more highly. *B*. State-dependent active adaptive management solution, given an initial uniform belief across models and a lower discount factor representing a decision maker who values short-term payoffs more highly.

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